

Syllabus - CMSI 3300.02 Artificial Intelligence - Fall 2024

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Course Information

- Course Name: CMSI 3300 (Section 02) Artificial Intelligence
- **Instructor**
  - Professor Lanyu Shang
  - Email: Lanyu.Shang@lmu.edu
  - Office: Doolan Hall 205
  - Office Hours: Tue/Thur 10 AM - Noon, Doolan Hall 205 or Zoom https://lmula.zoom.us/j/86471072428

- **Meeting Time and Location:**
  - Tues/Thurs 03:40 PM - 05:20 PM
  - **Seaver Hall Room 205 (New Location)**
- **Description:** CMSI 3300 is a 4-Unit course providing the introductory theory behind, and practice implementing, a wide swath of solutions to problems in the domain of artificial intelligence. Students will not only be able to match problems to an appropriate approach, but will understand the history, mathematics, and development of past approaches that have led the research and industry to its current state.

Learning Outcomes

By the course's end, students will:

- be introduced to the many active areas of research in artificial intelligence, including: logical inference, probabilistic & causal reasoning, fundamentals of data science, machine learning, deep learning, and generative models.
- understand existing approaches to a variety of classic and realistic AI problems, including those within: planning, reasoning under uncertainty, utility theory, value of information, reasoning over time, classification, and imitation learning.
- gain practice using popular data-structures in AI, including: knowledge bases, search trees, planning graphs, Bayesian Networks, decision networks, Naive Bayes Classifiers, regression and logistic regression, ensemble methods, and deep neural networks.
- become familiar with popular AI frameworks and libraries in Python (e.g., Numpy, Scikit, and PyTorch), and design artificial agents whose behaviors can be tangibly observed.
- grasp the ongoing avenues for research in the field, investigating some efforts that suit student-specific interests.

Prerequisites

Note: before taking this course, it is assumed that you have a matured grasp on the **Python** programming language, as well as the topics from Data Structures (CMSI 2120) and Algorithms (CMSI 2130). **We will spend little time in-class reviewing each!**

- **Data Structures:** although in Java for its code examples, you can consult the course archives for CMSI 2120 (from Professor Andrew Forney).
- **Algorithms:** the lecture of CMSI 2130 (from Professor Andrew Forney) have code-agnostic lessons on all of the needed concepts.
- **Python:** consult your CMSI 1010 instructor's materials or any number of online tutorials like 🔗Berkeley Python Crash Course. For practice, see our 🔗LMU CMSI Resources.
- **Git:** you should be comfortable with the fundamentals of Git and managing GitHub repositories as GitHub Classroom will be used to distribute and collect all assignments in the course. Need a refresher? See 🔗Git - Documentation.

Course Materials

We will be "using" one **optional** textbooks for the course, which will primarily cover or dive deeper into the theoretical material than we have the time to during lecture:

- [Textbook] Artificial Intelligence: A Modern Approach, ISBN 9780134610993 / 9789356063570

Although these specific texts are **not mandatory** for the course, they contain all concepts that we will be covering during the lectures in great detail.

Resources will be provided throughout the course detailing any additional topics that are not sufficiently contained in the above text. These resources will be free and publicly available.

Lecture Attendance

- **Not mandatory to attend:** lecture attendance is strongly advised since you'll be missing out on lecture content that may not be summarized in the course notes, like questions asked by peers, classwork exercises, and hints for homework and exams.

- **No need to inform the instructor of missed lecture:** I'll just assume you'll be making things up by reading online / asking friends.
- **Lectures are participatory:** come prepared to ask and answer questions to make the most out of your education!
- **Laptops discouraged, note-taking encouraged:** take notes by hand and summarize the important points -- don't copy verbatim.

Workload Expectations

- As this is a 4-Unit class, it is expected that you will be allocating an average of 9 hours outside of lectures per week working on course assignments, reading, and studying. You should manage your time appropriately.
- Some weeks will have less material to keep you busy, and others more, but be aware that this is a work-intensive course that may require you to spend a lot of time studying and programming!

Tentative Schedule of Topics

Item	Date	Topic
Lecture 1-1	8/27/2024	Introduction, Current Frontiers in AI
Lecture 1-2	8/29/2024	Propositional Logic
Lecture 2-1	9/3/2024	Propositional Inference
Lecture 2-2	9/5/2024	Propositional Inference
Lecture 3-1	9/10/2024	Intro to Probabilistic Reasoning
Lecture 3-2	9/12/2024	Probabilistic Reasoning
Lecture 4-1	9/17/2024	Bayesian Networks - Structure
Lecture 4-2	9/19/2024	Bayesian Networks - Semantics
Lecture 5-1	9/24/2024	Bayesian Networks - Inference
Lecture 5-2	9/26/2024	Bayesian Networks - Inference Optimizations
Lecture 6-1	10/1/2024	Bayesian Networks - Sampling
Lecture 6-2	10/3/2024	Decision and Utility Theory

Item	Date	Topic
Lecture 7-1	10/8/2024	Decision Networks
Lecture 7-2	10/10/2024	Value of Information
✏️ Midterm	10/15/2024	
Lecture 8-2	10/17/2024	Markov Models
Lecture 9-1	10/22/2024	Hidden Markov Models
Lecture 9-2	10/24/2024	Particle Filtering
Lecture 10-1	10/29/2024	Intro to Machine Learning and Data Science
Lecture 10-2	10/31/2024	Naive Bayes Classifiers
Lecture 11-1	11/5/2024	Tuning, Smoothing, and Optimization
Lecture 11-2	11/7/2024	Linear Perceptrons
Lecture 12-1	11/12/2024	Logistic Regression
Lecture 12-2	11/14/2024	Logistic Regression
Lecture 13-1	11/19/2024	Artificial Neural Networks
Lecture 14-1	11/21/2024	Intro to Deep Learning
Lecture 14-2	11/26/2024	Intro to Deep Lmitation Learning
🎉 Holiday	11/28/2024	Thanksgiving!
Lecture 15-1	12/3/2024	Intro to Generative Models
Lecture 15-2	12/5/2024	Current Topics in AI + Programming Lab
✏️ Final	Final Exam Schedule	

Assignments & Grading

Grade Decomposition

Grades will be assigned based on the following weighted coursework:

- **[25%] Classwork:** Small classwork assignments will be given roughly every 2 weeks, with in-class time allocated for their completion. Though you will have the opportunity to finish these in groups and with the instructor's assistance during class, you are not required to attend lectures, and so may submit these

electronically by their listed due dates. *Your lowest classwork grade will be dropped (even if you skipped it entirely)*.

- **[40%] Homework:** Larger assignments with heavy coding required. Programming style and comments are graded as well. All homework assignments are weighted equally. Expect roughly 5 assignments in total, apportioned once every ~2.5 weeks.
- **[35%] Exams:** There will be 2 exams (a midterm and a final). Your best exam score will constitute 2/3 of the exam grade, with your worse score constituting the remaining 1/3. For example, if your 2 exam scores were 60 and 90, then your weighted exam score for the course will be 80.0 (= 60*(1/3) + 90*(2/3))

Note: Because of the density of course content, classwork and homework exercises (but NOT exams) can be completed in **groups of up to 3 students.**

Final Grades

Final letter grades are given based on the university scale of grade percentages:

- A: 93 - 100
- A-: 90 - 92
- B+: 86 - 89
- B: 83 - 85
- B-: 80 - 82
- C+: 76 - 79
- C: 73 - 75
- C-: 70 - 72
- D: 65 - 69
- F: 64 and below

Fractional grade percentages at 0.5 or over will be rounded up, so an 89.5% will be considered a 90% (A-), but an 89.4% will be considered an 89% (B+) on the above scale.

That said, these are only the guaranteed grade assignments: if your "final" grade is an 82%, you are guaranteed a B- or better, but you might still get an A if 82% is the top score.

Extra Credit

Extra credit opportunities will be sporadically available during lectures / exams; if you are present and can hand in an attempt at the extra credit opportunities, you will receive bonus points!

These bonuses are applied post-difficulty-adjustments, so no student will be punished for not attending lectures relative to their peers.

Submission Standards

- **Assignment Posting:** all assignments (classwork, homework, or otherwise) will be announced in class and posted on Brightspace with due dates. You are responsible for attending the lectures / checking Brightspace/ your email for any updates to assignments and their associated deadlines.
- **Late Policy:** assignments are due at exactly the time indicated by the method specified. Assignments that are late by anywhere from 0 - 24 hours will only receive 80% of the points that they would have otherwise earned. Assignments that are late by 24+ hours receive a 0. There are **no exceptions** to this rule unless there are extenuating circumstances that have been discussed with and approved by the instructor prior to the deadline. Documentation may be required for any extensions granted.
- **Submission Errors:** "I forgot to push to GitHub" is not a valid excuse for late assignments; always double check your remote repository to make sure that your submission is present before the deadline. Moreover, assignments that contain syntax errors or deviate from given skeletons will receive a 0.
- **Style:** assignments which are sloppily submitted without proper formatting, style, and comments are subject to penalties. We will discuss what constitutes such offenses in the first classwork, though you should be familiar with clean coding standards from your introductory courses.

Academic Honesty

Students are encouraged to talk and think about problems in groups, but each submission should be their own, containing no code that has been copy-pasted from another student. We take plagiarism very seriously, and each submission will be run through a similarity-checking mechanism to ensure fairness. Code or homework which we believe to be shared innapropriately between students or copied from the internet is subject to severe disciplinary action (not that you would, just sayin'). The following are examples of **unacceptable** behavior in this course:

- Copying non-trivial amounts of code from the ANY other source into an assignment (excluding small things from StackOverflow like how to use Python list comprehension to simplify some code). This includes code found on the internet or through the use of a generative AI model like ChatGPT.
- Copying code from another student's submission, or having them verbally explain their submission line-by-line and you copying that.
- Copying another group's solutions on a classwork exercise (sharing between group members is encouraged... that's what it means to... you know... group up).

If you plan to copy anything from the internet, ask first!

The following CMSI Academic Honesty Standards provide more CMSI-specific standards for academic honesty; it is expected that by reading this syllabus you are aware of these standards.

Rule of thumb: if anyone but you is responsible for your submitted work, you are liable for academic dishonesty! Remember that it is YOU taking this course to learn the tools of your chosen trade, your degree will be useless otherwise!

Resources

Here are some LMU links for University Resources that might be of interest:

- Computer Science Department - Student Guide: There are some great review and practice guides in here too!
- Academic Degree Requirements and Policies: Pay particular attention to honor code and process.

- Disability Support Services (DSS)
- Student Codes and Policies
- Academic Resource and Learning Center

Acknowledgments

This course was originally designed and has been taught in the past few years by Professor Andrew Forney. I have followed his syllabus and course design closely. I appreciate his willingness to share the materials and for his contribution to the development of this course.